

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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- Disconnect the battery cables. See equipment manufacturer service information.
- Remove the alternator and alternator bracket. Refer to Procedure 013-001 in Section 13. (</qs3/pubsys2/xml/en/procedures/35/35-013-001-tr.html>)
- Remove the gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-031-tr.html>)
- Remove the accessory drive. Refer to Procedure 009-001 in Section 9. (</qs3/pubsys2/xml/en/procedures/35/35-009-001-tr.html>)
- Remove the hydraulic pump, if equipped. Refer to Procedure 009-016 in Section 9. (</qs3/pubsys2/xml/en/procedures/101/101-009-016-tr.html>)
- Disconnect the engine position wire connector, and remove the engine position sensor from the back side of the housing.
- Remove the water pump. Refer to Procedure 008-062 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-062-tr.html>)
- Remove the idler gears.
 - Refer to Procedure 001-036 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-036-tr.html>)
 - Refer to Procedure 001-039 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-039-tr.html>)

tr.html)

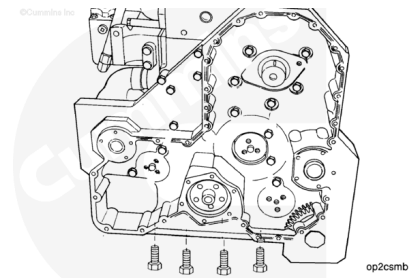
- Refer to Procedure 001-040 in Section 1.
(/qs3/pubsys2/xml/en/procedures/35/35-001-040-tr.html)

- Remove the camshaft gear or the camshaft and gear assembly.

- Refer to Procedure 001-012 in Section 1.
(/qs3/pubsys2/xml/en/procedures/35/35-001-012-tr.html)
- Refer to Procedure 001-008 in Section 1.
(/qs3/pubsys2/xml/en/procedures/35/35-001-008-tr.html)

Remove

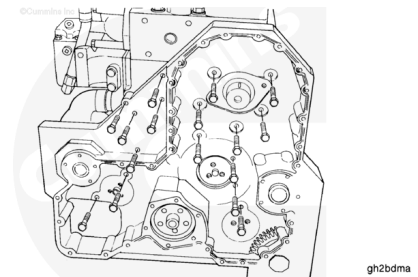
Remove the four oil pan-to-gear housing capscrews.



Remove the capscrews that secure the gear housing to the cylinder block.

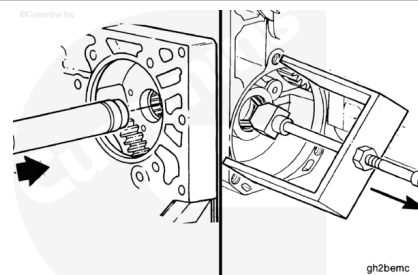
Note : Be careful not to damage the oil pan gasket when removing the gear housing. If the gasket is damaged, the oil pan must be removed and the gasket replaced.

Remove the gear housing from the cylinder block.



Use bearing driver kit, Part Number 3824117, to remove the water pump drive needle bearing.

Use bearing puller kit, Part Number 3823774, to remove the hydraulic drive needle bearing.

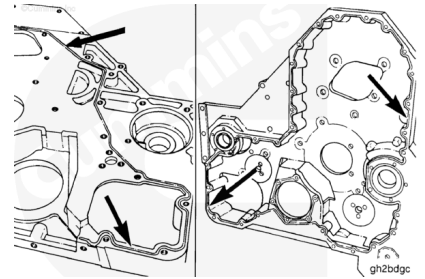


Clean and Inspect for Reuse

Remove the RTV sealant from the rear surface of the gear housing.

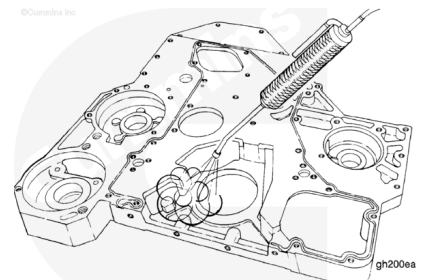
Remove the seal in the grooves on the front surface of the gear housing.

Use a wire brush, rounded blade, or screwdriver to clean the grooves on both sides.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

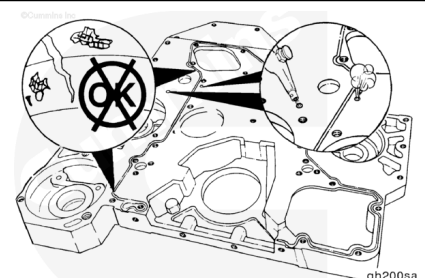
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use steam or solvent to clean the gear housing.

Dry the gear housing with compressed air.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



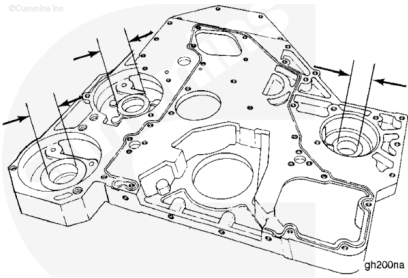
Inspect the gear housing for cracks and other damage.

Inspect the seal grooves on both sides of the housing for debris and damage.

Inspect the o-ring groove at the accessory drive oil jumper gallery for cracks and other damage.

Use compressed air to make sure the oil jumper gallery to the accessory drive is free of any obstructions.

Measure the inside diameter of the bearing bores at the accessory drive, water pump drive, and hydraulic drive.



Gear Housing Bearing Bore Inside Diameter			
	mm		in
Hydraulic Drive	41.967	MIN	1.6522
	41.992	MAX	1.6532
Water Pump	36.967	MIN	1.4553
	36.992	MAX	1.4564
Accessory Drive	45.100	MIN	1.7756
	45.125	MAX	1.7766

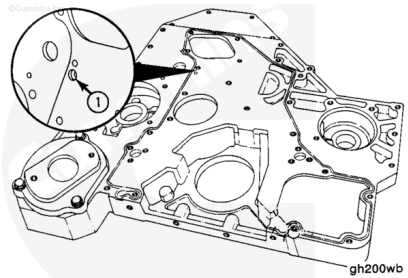
Install

⚠ CAUTION ⚠

Avoid the use of excessive amounts of sealant, which could result in blocked oil passages in the engine and cause engine damage.

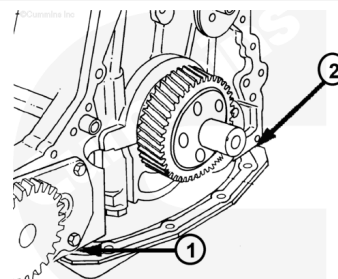
Install a new o-ring, Part Number 3883150, into the rear of the gear housing at the oil jumper gallery (1) for the accessory drive support. Be sure to keep sealant from the hole.

Note : Sealant requires assembly in 10 minutes or less. It is best to apply the sealant and then immediately assemble the parts.

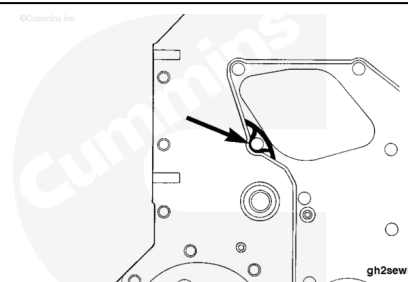


Apply a bead of sealant, Part Number 3164067, to the groove in the rear of the gear housing. Completely fill the groove so that approximately 1/16 to 1/8 inch of the bead is raised above the block mounting surface of the gear housing.

Apply a 2-mm [1/16-in] bead of sealant, Part Number 3164067, to the oil pan and cylinder block gear housing joint.



Apply additional sealant in the area by the camshaft opening, as shown in the illustration.



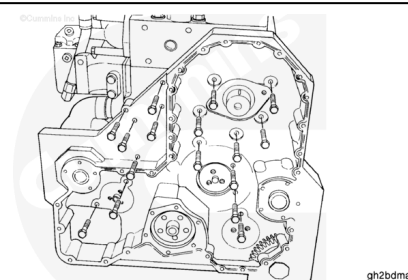
Install two guide pins, Part Number 3376488, in the gear housing end of the cylinder block.

Use care **not** to damage the oil pan gasket while installing the gear housing.

Install the gear housing and eleven capscrews (M10-1.50 x 25) in the inner part of the gear housing.

Install five capscrews (M10-1.50 x 50) on the outer surface of the housing.

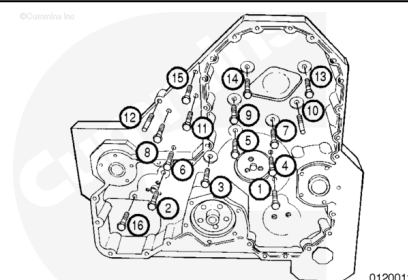
Remove the guide pins and install the two remaining capscrews (M10-1.50 x 25).



Tighten the capscrews in the sequence shown.

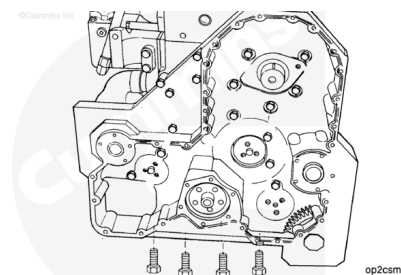
Torque Value:

1. 20 n•m [177 in-lb]
2. 68 n•m [50 ft-lb]



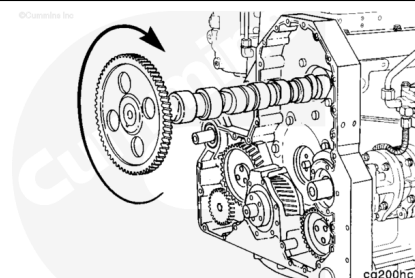
Install the four oil pan to gear housing capscrews.

Torque Value: 47 n·m [35 ft-lb]



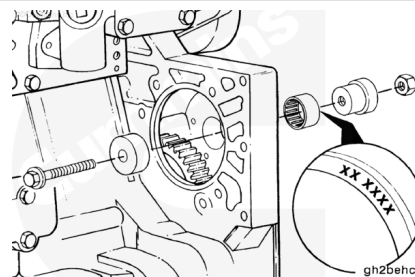
Install the camshaft gear, or camshaft and gear assembly.

- Refer to Procedure 001-012 in Section 1.
(/qs3/pubsys2/xml/en/procedures/35/35-001-012-tr.html)
- Refer to Procedure 001-008 in Section 1.
(/qs3/pubsys2/xml/en/procedures/35/35-001-008-tr.html)



⚠ CAUTION ⚠

The bearing must be installed with the part number side of the bearing against the installation tool to prevent damage to the bearing during the installation.



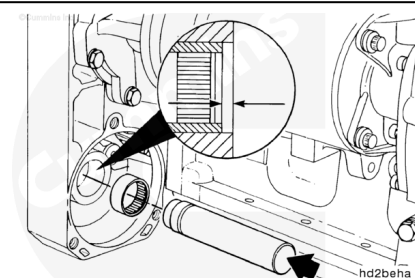
Use bearing driver kit, Part Number 3824117, to install a new needle bearing in the water pump drive bore of the gear housing.

Install the bearing from the front side of the gear housing, until the bearing is flush with the front edge of the housing bore.

Use bearing installation tool, Part Number 3823776, included in bearing puller kit, Part Number 3823774, to install a new needle bearing in the hydraulic drive bore of the gear housing.

Tap the bearing gently until it comes in contact with the shoulder in the housing.

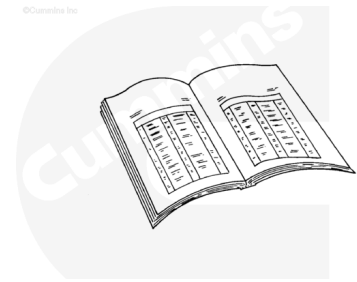
The bearing **must** be 0.25 to 0.76 mm [0.010 to 0.030 in] past the outside edge of the gear housing bore surface.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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- Install the idler gears.
 - Refer to Procedure 001-036 in Section 1.
(/qs3/pubsys2/xml/en/procedures/35/35-001-036-tr.html)
 - Refer to Procedure 001-039 in Section 1.
(/qs3/pubsys2/xml/en/procedures/35/35-001-039-tr.html)
 - Refer to Procedure 001-040 in Section 1.
(/qs3/pubsys2/xml/en/procedures/35/35-001-040-tr.html)
- Install the water pump. Refer to Procedure 008-062 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-062-tr.html)
- Install the engine position sensor on the back side of the housing. Connect the engine position wire connector.
- Install the hydraulic pump, if equipped. Refer to Procedure 009-016 in Section 9.
(/qs3/pubsys2/xml/en/procedures/101/101-009-016-tr.html)
- Install the accessory drive. Refer to Procedure 009-001 in Section 9. (/qs3/pubsys2/xml/en/procedures/35/35-009-001-tr.html)
- Install the gear cover. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/35/35-001-031-tr.html)
- Install the alternator and alternator bracket. Refer to Procedure 013-001 in Section 13.
(/qs3/pubsys2/xml/en/procedures/35/35-013-001-tr.html)
- Connect the battery cables. Refer to the OEM service manual.
- Operate the engine to normal operating temperature and check for leaks.

